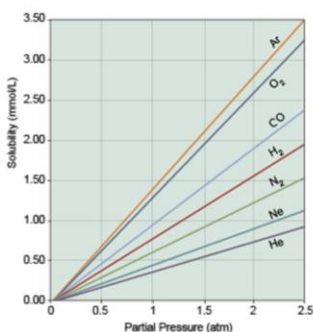


Gas solubility is affected by pressure

Pressure does not affect the solubility of liquids and solids.

The solubility of dissolved gas (C_g) is proportional to the partial pressure of that gas (P_g) above the solution.



Henry's Law

$$C_g = k P_g$$

k = Henry's Law constant



CHEM 110B - Lecture 32 PLA

1

The solubility of a gas in a solution is affected by the pressure. Pressure does not, however, affect the solubility of liquids and solids in solutions.

The solubility of a dissolved gas is proportional to the partial pressure of that gas (P_g) above the solution. This relationship is known as Henry's Law, which is shown on the slide as $C_g = k \cdot P_g$, where C_g is the concentration (i.e., solubility) of a gas ("g" is the identity of the gas), P_g is the partial pressure of the gas, and k is the proportionality constant. The proportionality constant (k) varies from one gas to the next, and its value is related to the strength of IMFs that the gas can form with the solvent. Gases that form stronger IMFs with water will have larger Henry's law constants, thus larger amounts of these gases will dissolve at a fixed partial pressure compared to weak IMF gases. In other words gases with stronger IMFs will be more soluble in water. This is shown by the graph on the bottom left of the slide. The graph plots solubility versus partial pressure of the gas, with each line representing a different gas. The slope of the line, which equals the Henry's law constant, is larger for gases that form stronger IMFs.

Henry's law explains why carbonated beverages release large amounts of gas when they're opened. These beverages are packaged with pressurized carbon dioxide, with a typical can of soda having an airspace consisting of CO₂ at a partial pressure of 1.5-2.0 atm. Under this high partial pressure, the CO₂ is quite soluble in the beverage. When the soda is opened, however, the airspace in the can (which was previously pressurized CO₂) is replaced with air at about 1 atm total pressure. Air only contains a tiny amount of CO₂, so the partial pressure of CO₂ goes from 1.5-2.0 atm to nearly zero. As such, the solubility of CO₂ in the soda decreases precipitously upon opening, resulting in a supersaturated solution. Since CO₂ is supersaturated in the opened soda we observe a net movement (in the form of bubbles and fizz) of gas out of the beverage.